Spatial-Semantic Alignment and Change-Aware Network for Remote Sensing Image Change Captioning

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***Abstract*—Remote sensing image change captioning aims to generate natural language descriptions of dynamic changes between bitemporal images. This task faces challenges of aligning high-level semantics with sparse change regions and modeling long-range dependencies in diverse change patterns. To alleviate these limitations, we propose the spatial-semantic alignment and change-aware network, dubbed SACNet. It introduces a semantic-spatial collaborative encoding module synergizing a CNN-based spatial encoder with a CLIP-based semantic encoder, using precise spatial cues to refine global semantic features. A change-driven adaptive feature calibration module emphasizes discriminative change signals via intertemporal feature dissimilarity modeling, suppressing background noise and highlighting key changes. In addition, a multipath state space model (SSM)-guided spatial representation module is designed to capture diverse complex spatial change topologies. Extensive experiments on the LEVIR-CC and WHU-CDC datasets demonstrate that SACNet achieves state-of-the-art performance, generating both high-precision textual descriptions and approximate change masks. Code will be available at https://github.com/CVer-Yang/SACNet**

***Index Terms*—Change captioning, key change features, multitask collaborative optimization, remote sensing, state space models (SSMs).**

I. Introduction

With the rapid advancement of high-resolution satellite [1], [2], [3] and remotely piloted aerial remote sensing [4], bitemporal imagery [5] has become readily available, providing unprecedented capabilities for monitoring dynamic Earth surface changes across urban development [6], ecological evolution [7], [8], [9], and disaster response [10], [11]. Traditional change detection methods output binary masks that only indicate where changes occur, lacking semantic interpretability, which leads to heavy reliance on expert analysis and hinders usability for nonspecialist decision-makers. To bridge this gap, remote sensing image change captioning (RSICC) has emerged as a cross-modal task that translates pixel-level bitemporal differences into human-understandable natural language descriptions, answering not only where but also what and how changes happened.

As a novel cross-modal task in remote sensing [8], remote sensing image change captioning emerges as a natural integration of remote sensing image captioning (RSIC) [12] and change detection [13]. Specifically, RSIC focuses on converting the visual content of single-temporal remote sensing images into natural language descriptions, with core emphasis on the semantic expression of individual scenes. For example, Yang et al. [14] propose an HCNet framework based on CNN-LSTM to describe ground object types and their relative positional relationships in remote sensing images. Wang et al. [15] further integrate object segmentation results to enhance alignment between descriptive texts and object locations. On the other hand, the change detection task concentrates on identifying pixel-level differences between bitemporal images and outputting binary change masks, aiming solely to answer whether and where changes occur [13], [16]. By fusing the fine-grained difference identification capability of change detection with the semantic description function of RSIC, RSICC is designed to achieve a complete semantic transformation from the spatial location of changes to the specific semantic content of changes, which is of great significance for urban monitoring, ecological evolution analysis, and disaster response [17], [18]. To advance this research direction, Liu et al. [17] introduce the LEVIR-CC dataset, the first benchmark for RSICC, which provides bitemporal remote sensing image pairs accompanied by manually annotated change masks and natural language descriptions. Subsequently, datasets like WHU-CDC [18] further enrich complex urban change research scenarios and provide critical support for method evaluation and cross-dataset generalization verification.

Despite recent progress, the generation of accurate change captions from high-resolution data remains a formidable challenge. Existing methodologies generally follow three technical paradigms, yet each faces distinct limitations that exacerbate two core dilemmas: the semantic-spatial dichotomy and

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the efficiency-richness tradeoff. **First, the semantic-spatial dichotomy.** Traditional RSICC frameworks [17], [19] rely on vanilla CNN or Transformer encoders to extract bitemporal features. These methods often treat remote sensing images as generic natural scenes and fail to effectively decouple sparse, discrete change signals from dominant, redundant static backgrounds. This limitation results in severe background interference, where models prioritize invariant objects over actual changes. Recently, foundation model adaptations (e.g., CLIP [20] and SAM [21]) have been introduced to provide strong cross-modal semantic priors [22], [23]. Nevertheless, these models typically operate on fixed-scale patches, which leads to *spatial indeterminacy* when localized change regions require fine-grained boundary perception. Consequently, these paradigms yield suboptimal captions: they are either spatially grounded but semantically shallow, or semantically plausible yet factually misaligned with the actual change locations [22], [24]. **Second, the efficiency-richness tradeoff.** The description of complex geographic evolution, such as linear road extensions or radial urban sprawl, necessitates the modeling of long-range spatial dependencies. While standard Transformers achieve global receptive fields, their quadratic computational complexity $\mathcal{O}(N^2)$ renders them computationally prohibitive for high-resolution remote sensing inputs [25]. To alleviate this, efficient local operators (e.g., depthwise convolutions) have been employed, but they sacrifice the global context essential for capturing cross-scene spatial correlations. Emerging state space models (SSMs), such as Mamba-based architectures [16], [26], offer a promising direction with linear complexity. However, current SSM-based RSICC methods predominantly utilize single-path scanning mechanisms. Such designs are inherently limited in characterizing the *anisotropic spatial topologies* of diverse geographic objects. Therefore, these approaches fail to bridge the gap between efficient computation and the rich representation required for coherent change description.

To alleviate these long-standing problems in RSICC, we propose a spatial-semantic alignment and change-aware network for RSICC, dubbed as SACNet, which unifies spatial precision, semantic richness, and computational efficiency through three core innovations tailored for the unique characteristics of remote sensing change scenes: 1) a *semantic-spatial collaborative encoding* module that injects multiscale CNN-derived positional cues derived from bitemporal feature differences into CLIP global semantic features, realizing precise spatial grounding of semantic representations and mitigating the semantic-spatial dichotomy; 2) a *change-driven adaptive feature calibration* mechanism that dynamically suppresses redundant background clutter via cross-temporal feature similarity inversion, amplifying discriminative change signals and focusing the model on meaningful change regions; and 3) the *multipath SSM-guided spatial representation module* specially designed for RSICC, which captures diverse complex spatial change topologies through parallel Mamba streams with multiple scanning paths while maintaining linear computational complexity, resolving the efficiency-richness tradeoff. The main innovations of this article are summarized as follows.

1) A *semantic-spatial collaborative encoding module* is proposed to alleviate the semantic-spatial dichotomy in RSICC. It fuses CLIP-derived global semantic features with CNN-extracted multiscale spatial change cues for bitemporal remote sensing imagery, while preserving cross-modal semantic consistency and fine-grained spatial localization for sparse change regions. Complementarily, a *change-driven adaptive feature calibration module* mitigates severe background interference in complex RSICC scenes by dynamically enhancing salient change regions and suppressing redundant static clutter via similarity-based recalibration, thus guiding the model to focus on task-relevant meaningful change features in remote sensing change analysis.
2) A *multipath SSM-guided spatial representation module* is established in bitemporal change modeling. By leveraging parallel Mamba streams with row, column, diagonal, and anti-diagonal scanning paths, this module captures diverse spatial topologies, including linear, clustered, and radial types with linear computational complexity, achieving both efficiency and expressive power.
3) A unified joint learning framework is developed for RSICC. It leverages pixel-level change masks and semantic-level captions as dual supervision signals, generates descriptions accurately reflecting change content, and yields satisfactory results on two widely used RSICC datasets.

The rest of this article is organized as follows. Section II provides related work in remote sensing image change captioning. In Section III, we describe the details of our proposed SACNet. The experimental results are analyzed in Section IV. Finally, Section V offers the conclusion.

II. Related Work

Current research on remote sensing image change captioning revolves around three core technical challenges: extracting discriminative change features precisely, modeling spatiotemporal dependencies in bitemporal data efficiently, and aligning visual change patterns with textual semantics robustly.

A. Encoder–Decoder-Based Bitemporal Feature Fusion Methods

Early RSICC methods build on classic image captioning frameworks to form a feature extraction-fusion-language generation pipeline, focusing on effective bitemporal feature representation and fusion. Liu et al. [17] propose the first CNN-Transformer-based RSICC framework that extracts visual features via CNN, fuses change mask information, and feeds results into a Transformer decoder for caption generation. This work lays the technical foundation but suffers from insufficient visual-linguistic alignment and inaccurate focus on change regions. Subsequent studies optimize feature extraction and fusion mechanisms. Chang and Ghamisi [19] present Chg2Cap with a dual-branch CNN and hierarchical self-attention fusion encoder, enabling change region localization without mask supervision and robustness to illumination or

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seasonal disturbances. Liu et al. [27] treat bitemporal images as a two-frame short video, using pretrained video encoders for spatiotemporal feature fusion and linear projection to language model space. Zhou et al. [28] propose a single-stream extractor network that concatenates bitemporal images into a six-channel input for contrastively pretrained ResNet feature extraction, explicitly modeling temporal differences via shallow embedding and cross-attention guided modules to alleviate domain shift and reduce computation.

These methods advance bitemporal feature fusion and change modeling within conventional encoder–decoder pipelines, achieving reliable change localization and robustness to complex environmental variations. However, they lack large-scale pretrained semantic priors and fail to model the long-range structured dependencies of geographic change events, leading to shallow semantic expression of generated captions for complex change scenarios.

B. Foundation Model Adaptation and Semantic Enhancement-Based Methods

Advancements in vision-language foundation models [29] and efficient modeling drive RSICC beyond traditional paradigms, with innovations centered on foundation model adaptation and semantic enhancement. Researchers leverage cross-modal advantages of pretrained models through remote sensing adaptation and semantic injection. Wang et al. [22] propose a SAM-guided framework that extracts region-level representations and integrates knowledge graphs to improve change recognition and caption semantic accuracy via cross-attention. Wang et al. [30] construct the large-scale RSICC dataset for two-stage training of visual encoders and large language models, designing a difference-aware fusion strategy to inject change semantics for subtle and complex change perception. Zhu et al. [23] propose Change3D that treats bitemporal images as mini-videos, inserting learnable perception frames for video encoder-based spatiotemporal modeling to unify change detection and captioning with fewer parameters. Yang et al. [31] realize pixel-level change detection and instruction-tuned LLM collaboration to enhance key change features and suppress pseudo-change interference. Hang et al. [24] integrate textual semantics into change feature extraction via text guidance to improve fine-grained description accuracy.

While foundation models like CLIP and SAM bring strong semantic and spatial priors, their adaptation for RSICC remains superficial. Lacking fine-tuning for bitemporal comparison, they suffer from spatial indeterminacy or high computational cost without adaptive filtering, and fail to explicitly model long-range structured dependencies in geographic changes.

C. Efficient Spatiotemporal Modeling and Multitask Optimization-Based Methods

Efficient spatiotemporal modeling and lightweight network design have become key research directions in RSICC, motivated by the need to mitigate the high computational complexity of Transformers. Among emerging techniques, SSMs have emerged as a promising paradigm. Their linear computational complexity and inherent global receptive field are properties well-suited for processing high-resolution remote sensing data.

SSM has evolved from remote sensing change detection to RSICC with gradual improvements, yet notable limitations. SPRMamba [16] introduced Mamba into change detection, but only predicts binary masks and lacks semantic capabilities, making it unsuitable for RSICC. RSCaMa [26] first applies Mamba to RSICC but adopts a single horizontal scanning path that only models linear dependencies and fails to capture complex spatial topologies. It does not utilize the multiscan SSM strategy, making it incapable of modeling the anisotropic geographic structures such as radial urban sprawl and diagonal road extension in remote sensing imagery. Beyond single-path scanning, existing multiscan SSM architectures are not tailored for RSICC tasks, such as VMamba [32], as they lack dedicated change-aware interaction mechanisms for bitemporal data and cannot effectively focus on sparse change regions in remote sensing imagery. Other efficient methods further enrich the RSICC technical system. Li et al. [33] propose TISDNet that realizes bidirectional bitemporal feature interaction via cross-temporal attention, designing a symmetric difference Transformer and bidirectional triplet ranking loss to improve semantic alignment. Liu et al. [34] propose a prompt learning-based decoupled paradigm that splits change classification and caption generation to reduce learning difficulty. Ferrod et al. [35] combine contrastive and multitask learning for image–text retrieval and change captioning to enhance cross-modal alignment. Zhu et al. [36] present semantic-CC that integrates foundation model knowledge via three-stage training for stable multitask collaboration. Liu et al. [37] propose a change-agent, integrating LLMs to construct interactive agents for diverse and personalized change caption generation.

Existing efficient spatiotemporal modeling methods still fail to resolve two core issues simultaneously: the semantic-spatial dichotomy and the tradeoff between computational feasibility and descriptive richness. SSM-based methods are constrained by single-path designs, which hinder the capture of diverse spatial topologies, ultimately limiting their effectiveness in describing complex geographic change events.

III. Proposed Method

A. Overview

As illustrated in Fig. 1, the proposed SACNet framework unifies semantic understanding and spatial reasoning in an end-to-end pipeline. The process starts with a dual-stream feature extractor where a CLIP and a ResNet capture global semantic features and multiscale spatial features from bitemporal images, respectively. These features undergo processing through three core innovative components. First, the semantic-spatial collaborative encoding module fuses semantic features of CLIP with CNN-derived spatial cues to generate spatially precise and semantically grounded feature representations. Subsequently, the change-driven adaptive feature calibration module refines these features by computing intertemporal dissimilarity, dynamically amplifying salient change signals

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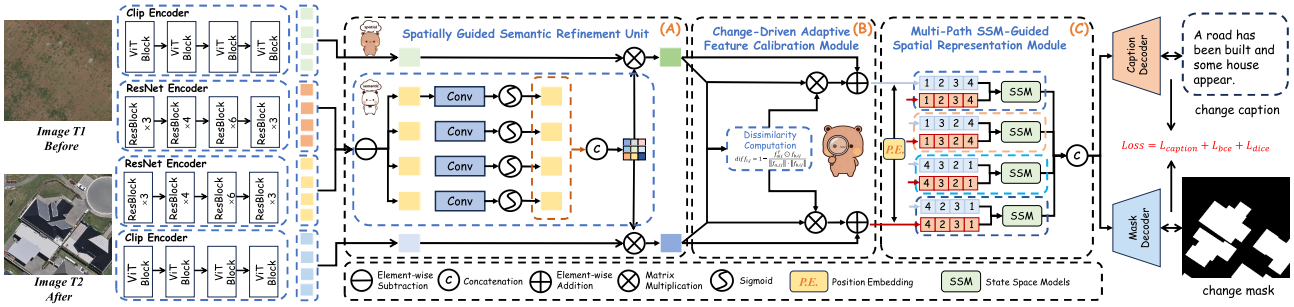


Fig. 1. Overview of the proposed SACNet for remote sensing image change captioning. (a) Spatial-semantic collaborative encoding fuses CNN-derived multiscale spatial cues with CLIP-based global semantic features, enhancing the correlation between change regions and semantic information to mitigate spatial indeterminacy. (b) Change-driven adaptive feature calibration emphasizes discriminative change features by modeling intertemporal feature dissimilarity, effectively suppressing background clutter while amplifying salient change signals. (c) Multipath SSM-guided spatial representation module uses SSM to fuse features for diverse change pattern modeling. A dual decoder generates change captions and masks, with joint caption-segmentation loss ensuring multitask consistency.

while suppressing background interference. To model diverse spatial topologies of the refined features, the multipath SSM-guided spatial representation module leverages parallel Mamba streams along different scanning paths to capture long-range dependencies. The resulting fused representation is fed into a dual-branch decoder that jointly predicts change masks and generates change captions.

B. Semantic-Spatial Collaborative Encoding Module

Visual feature encoding is critical for RSICC, which demands semantic consistency of bitemporal features, accurate change region localization, and effective bitemporal difference characterization. Traditional single-modal encoding schemes exhibit inherent limitations for RSICC tasks. CLIP-based encoding delivers strong semantic generalization yet lacks fine-grained spatial perception and is unoptimized for bitemporal differences, while CNN-based encoding excels at spatial localization but suffers from weak semantic expression. To alleviate this, we propose the semantic-spatial collaborative encoding module tailored for RSICC. This module generates dynamic change-aware masks from CNN-extracted multiscale spatial features, enhancing change signals and suppressing backgrounds in CLIP semantics via mask weighting to realize spatial-semantic fusion and precise bitemporal change embedding. It comprises three components: a CLIP-based semantic encoder, a CNN-based multiscale spatial extractor, and a spatially guided semantic refinement (SGSR) unit. Detailed implementation details are provided below.

1) *CLIP Semantic Encoding Branch:* We adopt a pretrained ViT-B/32 CLIP model as the basic semantic encoder, which constructs a unified cross-modal semantic space via large-scale image–text contrastive learning. A parallel encoding paradigm is designed for bitemporal inputs to synchronously encode I_A and I_B, yielding high-dimensional semantic features $feat_A, feat_B$ and global semantic embeddings $clip_A, clip_B$. CLIP pretrained cross-modal space ensures natural semantic consistency and alignment of bitemporal features, providing a consistent semantic benchmark for subsequent bitemporal difference interaction.
2) *CNN Spatial Encoding Branch:* We adopt ResNet-50 as the CNN backbone with a parallel bitemporal strategy to extract multiscale spatial features E_1, E_2, E_3, E_4. This branch provides hierarchical spatial cues for both temporal views, compensating for CLIP's weak spatial localization and establishing a multiscale reference for bitemporal differencing to precisely characterize change regions.
3) *SGSR Unit:* Spatially aware semantic refinement for bitemporal images is achieved through a guidance unit that modulates CLIP-derived semantic representations using multiscale CNN features. Given the CNN feature pyramids $\{E_{A,i}\}_{i=1}^4$ and $\{E_{B,i}\}_{i=1}^4$ extracted from the two temporal views, absolute differences are computed at each scale to characterize spatial discrepancies, i.e.,

$$\text{diff}_i = |E_{A,i} - E_{B,i}| \tag{1}$$

$$\text{mask}_i = \sigma\left(\text{Conv1} \times 1\left(\text{diff}_i\right)\right) \tag{2}$$

where $E_{A,i}$ and $E_{B,i}$ denote the CNN features at scale $i \in \{1, 2, 3, 4\}$, and $\sigma(\cdot)$ is the Sigmoid activation function. These operations yield initial change response masks at multiple scales for preliminary localization of change regions.

Complementary cues from shallow and deep features are integrated by upsampling all mask_i to the spatial resolution of $E_{A,1}$ via bilinear interpolation, where shallow layers provide precise spatial localization and deeper layers capture rich semantic context. The aligned masks are subsequently fused through a 3×3 convolution to produce a globally consistent refined mask, i.e.,

$$\text{mask}_{\text{fused}} = \text{Conv3} \times 3\left([\uparrow \text{mask}_1;\ \uparrow \text{mask}_2;\ \uparrow \text{mask}_3;\ \uparrow \text{mask}_4]\right) \tag{3}$$

where $[\cdot]$ denotes channelwise concatenation and $\uparrow$ denotes bilinear upsampling.

Alignment with the 7×7 patch grid of CLIP features is accomplished by interpolating the fused mask and flattening it. The resulting flattened mask is applied to the bitemporal CLIP semantic features $F_A, F_B \in \mathbb{R}^{B \times 49 \times D}$ via elementwise multiplication to enhance change-related semantics. An identity-preserving residual branch, denoted res$(\cdot)$, is introduced to prevent loss of original semantic information during enhancement. This branch is implemented as a linear

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transformation with weights initialized to the identity matrix and bias set to zero. The optimized features are thus expressed as

$$F_A^{\text{opt}} = (F_A \odot \text{mask}_{\text{flat}}) + \text{res}(F_A) \tag{4}$$

$$F_B^{\text{opt}} = (F_B \odot \text{mask}_{\text{flat}}) + \text{res}(F_B) \tag{5}$$

where $\odot$ denotes elementwise multiplication. This mechanism dynamically suppresses background regions in CLIP features using CNN-derived spatial change cues while simultaneously enhancing semantic representations in changed regions across both temporal views, thereby achieving spatially grounded semantic refinement for bitemporal image understanding.

C. Change-Driven Adaptive Feature Calibration Module

Most existing methods perform subtraction operations on global features. Such an approach fails to adaptively distinguish change regions from complex background clutter in remote sensing imagery, which dilutes discriminative change cues and leads to substantial performance degradation in challenging scenarios. To alleviate this problem, we propose the change-driven adaptive feature calibration module with adaptive feature modulation. This module is tailored to address core RSICC challenges: cross-temporal feature distribution shifts from illumination variations and sensor discrepancies, excessive background redundancy, and pseudo-change interference due to sparse change regions.

In implementation, layer normalization is applied to the bitemporal change features F_A^{opt} and F_B^{opt} to mitigate distribution shifts induced by illumination variations and sensor discrepancies, thereby ensuring reliable difference computation, i.e.,

$$\hat{F}_A = \text{Reshape}\left(F_A^{\text{opt}}\right) \tag{6}$$

$$\hat{F}_B = \text{Reshape}\left(F_B^{\text{opt}}\right) \tag{7}$$

where Reshape$(\cdot)$ denotes the format transformation between a 1-D sequence and a 2-D feature map.

Subsequently, pixelwise change saliency is quantified via cosine similarity. Specifically, the similarity between normalized features at spatial location (i, j) is given by

$$S_{i,j} = \frac{\hat{F}_A[:, i, j] \cdot \hat{F}_B[:, i, j]}{\left\|\hat{F}_A[:, i, j]\right\|_2 \left\|\hat{F}_B[:, i, j]\right\|_2 + \varepsilon} \tag{8}$$

where $\varepsilon = 10^{-8}$ prevents numerical instability. Since high similarity implies semantic consistency, a change response mask is obtained by inverting and smoothing the similarity map through a Sigmoid function, i.e.,

$$M_{\text{change}} = 1 - \sigma(S) \tag{9}$$

with $\sigma(\cdot)$ denoting the Sigmoid activation. The resulting mask $M_{\text{change}} \in \mathbb{R}^{B\times 1\times H'\times W'}$ assigns higher values to regions exhibiting significant change. Finally, adaptive feature refinement is performed by modulating the normalized features with the change mask. Elementwise multiplication enhances change-related signals while attenuating background responses, i.e.,

$$F_A^{\text{enhance}} = \text{IReshape}\left(\hat{F}_A \odot \left(1 + M_{\text{change}}\right)\right) \tag{10}$$

$$F_B^{\text{enhance}} = \text{IReshape}\left(\hat{F}_B \odot \left(1 + M_{\text{change}}\right)\right) \tag{11}$$

where $\odot$ denotes elementwise multiplication, and IReshape$(\cdot)$ is the inverse transformation of Reshape$(\cdot)$, converting 2-D feature map back to 1-D sequence.

D. Multipath SSM-Guided Spatial Representation Module

Effective change captioning requires modeling local changes and spatial configurations. These diverse topological patterns cannot be captured by a single sequential ordering. The SSM achieves linear complexity and global receptive field via adaptive parameters, enabling efficient long-range dependence modeling. However, its unidirectional processing cannot characterize heterogeneous spatial dependencies in bitemporal remote sensing images. Inspired by VMamba, we propose the multipath SSM-guided spatial representation module, which encodes multiple spatial traversal patterns into parallel SSM streams. It captures rich cross-spatial change relationships through three steps: change-aware interaction, multipath spatial fusion, and feature update.

As illustrated in Fig. 2, we introduce learnable row and column positional embeddings to compensate for the spatial indeterminacy of flattened sequence features F_A, F_B. The spatially augmented features are obtained by adding the flattened positional embedding PE to the input features, i.e.,

$$F_A^{\text{spatial}} = F_A + \text{PE} \tag{12}$$

$$F_B^{\text{spatial}} = F_B + \text{PE}. \tag{13}$$

To facilitate deep interaction between the two temporal streams, we compute an explicit change signal via feature differencing as a discriminative prior for change-aware dependence modeling, i.e.,

$$\Delta F = F_B^{\text{spatial}} - F_A^{\text{spatial}}. \tag{14}$$

The change-aware interaction between each temporal feature stream and the change signal is encapsulated as an operation termed InterMamba. Distinct from the canonical single-input Mamba that solely processes individual sequence data, this tailored operation integrates spatial difference-aware characterization to boost fine-grained change perception capability and capture long-range spatial dependencies simultaneously, which can be simply denoted as the following interactive formulation for intuitive expression, i.e.,

$$F_A^{\text{interact}} = \text{InterMamba}\left(F_A^{\text{spatial}}, \Delta F\right) \tag{15}$$

$$F_B^{\text{interact}} = \text{InterMamba}\left(F_B^{\text{spatial}}, \Delta F\right). \tag{16}$$

Notably, the InterMamba operation follows a spatial difference-aware, Mamba-driven interaction pipeline; the complete implementation mechanism of this operation is elaborated step by step as follows.

For each temporal stream feature, we perform preprocessing including layer normalization, linear projection, depthwise convolution, and SiLU activation to capture local spatial correlations and stabilize feature distribution, i.e.,

$$F_A^p = \text{SiLU}\left(\text{DWC}\left(\text{Linear}\left(\text{LayerNorm}\left(F_A^{\text{spatial}}\right)\right)\right)\right) \tag{17}$$

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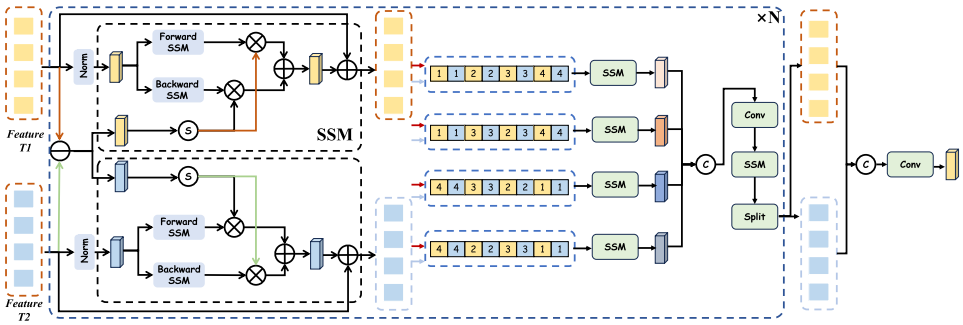


Fig. 2. Overview of the proposed multipath SSM-guided spatial representation module.

$$F_B^p = \mathrm{SiLU}\left(\mathrm{DWC}\left(\mathrm{Linear}\left(\mathrm{LayerNorm}\left(F_B^{\mathrm{spatial}}\right)\right)\right)\right) \quad (18)$$

where DWC(·) denotes depthwise convolution. We then adopt bidirectional state space model scanning to alleviate the unidirectional limitation of standard Mamba, enabling comprehensive spatial dependence modeling, i.e.,

$$F_A^f = \mathrm{SSM}\left(F_A^p\right) \quad (19)$$

$$F_A^b = \mathrm{flip}\left(\mathrm{SSM}\left(\mathrm{flip}\left(F_A^p\right)\right)\right) \quad (20)$$

$$F_B^f = \mathrm{SSM}\left(F_B^p\right) \quad (21)$$

$$F_B^b = \mathrm{flip}\left(\mathrm{SSM}\left(\mathrm{flip}\left(F_B^p\right)\right)\right) \quad (22)$$

where flip(·) denotes sequence reversal and SSM(·) represents the SSM processing. The change signal is processed to generate a change-aware weight map; this weight map modulates the bidirectional SSM outputs to enhance focus on change-related regions, i.e.,

$$F_d = \mathrm{SiLU}\left(\mathrm{Linear}\left(\Delta F\right)\right). \quad (23)$$

The interacted features are obtained by linear projection of the modulated bidirectional outputs and residual connection with the original spatial features, preserving the initial spatial information while integrating change-aware dependencies, i.e.,

$$F_A^{\mathrm{interact}} = \mathrm{Linear}\left(F_A^f \odot F_d + F_A^b \odot F_d\right) + F_A^{\mathrm{spatial}} \quad (24)$$

$$F_B^{\mathrm{interact}} = \mathrm{Linear}\left(F_B^f \odot F_d + F_B^b \odot F_d\right) + F_B^{\mathrm{spatial}} \quad (25)$$

where $\odot$ denotes elementwise multiplication.

To model complex anisotropic spatial structures that cannot be captured by a single traversal order, we propose a multipath spatial fusion strategy. First, the interacted features are normalized via layer normalization, and the normalized features of the two temporal streams are concatenated along the channel dimension to form a unified bi-temporal feature tensor $F_{\mathrm{fuse}} = \mathrm{Concat}(F_A^{\mathrm{interact}}, F_B^{\mathrm{interact}})$. We then design four distinct spatial traversal strategies, horizontal, vertical, reverse horizontal, and diagonal traversal, denoted as $\{\mathrm{Scan}_k\}_{k=1}^4$. Each strategy reshapes the flattened sequence back to a 2-D grid and reorders the spatial dimensions to form a unique 1-D sequence, enabling the model to capture spatial dependencies along specific axes. Each path-specific sequence is processed by a dedicated parallel InterMamba block, leveraging linear complexity and the global receptive field of Mamba to model long-range dependencies along the traversal direction as

$$F_{\mathrm{path}}^{proc,k} = \mathrm{InterMamba}\left(F_{\mathrm{path}}^{seq,k}, F_{\mathrm{path}}^{seq,k}\right) \quad (26)$$

where $F_{\mathrm{path}}^{seq,k}$ denotes the sequence of the kth traversal path. The processed features from the four paths are aggregated via channelwise concatenation and linear projection to compress the feature dimension back to the target dimension consistent with F_{fuse}

$$F_{\mathrm{agg}} = \mathrm{Concat}\left(F_{\mathrm{path}}^{proc,1}, F_{\mathrm{path}}^{proc,2}, F_{\mathrm{path}}^{proc,3}, F_{\mathrm{path}}^{proc,4}\right). \quad (27)$$

A final fusion block aggregates the multipath contextual information into a unified bi-temporal representation, integrating diverse spatial dependencies into a coherent feature space, i.e.,

$$F_{\mathrm{fused}} = \mathrm{InterMamba}\left(F_{\mathrm{agg}}, F_{\mathrm{agg}}\right). \quad (28)$$

The fused bi-temporal representation is split back into two single-stream features corresponding to the two temporal snapshots

$$\hat{F}_A, \hat{F}_B = \mathrm{Split}\left(F_{\mathrm{fused}}\right). \quad (29)$$

We introduce a learnable weighted residual connection to preserve original change-aware information and stabilize training, adaptively balancing the contribution of the multipath fused feature and the normalized interacted feature

$$F_A^{\mathrm{update}} = \hat{F}_A + \alpha \cdot \mathrm{LayerNorm}\left(F_A^{\mathrm{interact}}\right) \quad (30)$$

$$F_B^{\mathrm{update}} = \hat{F}_B + \alpha \cdot \mathrm{LayerNorm}\left(F_B^{\mathrm{interact}}\right) \quad (31)$$

where α is a learnable scalar parameter. After the single-layer interaction and fusion steps, the updated features F_A^{update} and F_B^{update} are reshaped back to 2-D feature maps in channel-first format and concatenated along the channel dimension. A 1×1 convolution projects the concatenated feature to the target dimension for the subsequent language decoder, followed by layer normalization and residual connection to enhance feature discriminability. The final feature F_{final} integrates change-aware bi-temporal interaction and multipath spatial context, providing a comprehensive representation for change caption generation.

E. Dual Decoder

The caption decoder adopts a standard Transformer decoder architecture with multihead masked self-attention, encoder–decoder attention, and feed-forward network sublayers. For encoder–decoder attention linear projections derive keys and values from the final fused feature F_{final} while queries are obtained from the preceding masked self-attention sublayer output to enable adaptive alignment between visual change

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features and linguistic tokens. The decoder output is fed into a linear layer followed by softmax activation to generate word probability distributions supporting autoregressive, semantically accurate change descriptions.

The mask decoder reshapes F_{final} into a 2-D feature map after a linear layer-based dimension adaptation. It employs four cascaded transposed convolution blocks for progressive upsampling, interleaved with fusion convolutions, batch normalization, and ReLU activation to refine features. The final stage maps features to a single-channel logit map via transposed convolution and 1×1 convolution, which is bilinearly interpolated to the target size for change mask generation.

F. Joint Loss Function

The loss function for the change detection task is L_{mask}, which adopts DiceBCELoss to alleviate the class imbalance problem. It consists of a combination of Dice loss and binary cross-entropy loss

$$L_{\text{mask}} = \text{Dice}\left(M_{\text{pred}}, M_{\text{gt}}\right) + \text{BCE}\left(M_{\text{pred}}, M_{\text{gt}}\right). \tag{32}$$

The semantic description task aims to generate a fluent textual caption reflecting the change semantics between bitemporal images. Let $\mathbf{s}^* = (s_1^*, s_2^*, \ldots, s_T^*)$ be the ground-truth (GT) token sequence. During training, we employ the teacher-forcing strategy, where the model predicts the current token by conditioning on the bitemporal visual context $\mathcal{I}_{\text{bitemp}}$ and the preceding GT tokens $s_{1:t-1}^*$. The captioning loss is formulated as the negative log-likelihood of the GT sequence under the model distribution, i.e.,

$$\mathcal{L}_{\text{text}} = -\sum_{t=1}^{T} \log p_\theta \left(s_t^* \mid \underbrace{s_{1:t-1}^*}_{\substack{\text{Textual} \\ \text{Context}}}, \underbrace{\mathcal{I}_{\text{bitemp}}}_{\substack{\text{Visual} \\ \text{Change}}} \right) \tag{33}$$

where $p_\theta(\cdot)$ represents the output probability of the language head. This is equivalent to the standard cross-entropy loss between predicted and GT token distributions at each decoding step.

The total training objective combines the change mask generation loss $\mathcal{L}_{\text{mask}}$ and the captioning loss $\mathcal{L}_{\text{text}}$ via time-dependent weighting coefficients, i.e.,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{text}}. \tag{34}$$

IV. Experiments

A. Datasets and Settings

Two mainstream change captioning datasets are adopted for experimental verification. Both datasets provide paired bitemporal images, pixel-level change masks, and corresponding change description texts. Following default experimental settings, the LEVIR-CC dataset is partitioned into 6815 training pairs, 1333 validation pairs, and 1929 testing pairs. The WHU-CDC dataset is divided into 5947 training pairs, 743 validation pairs, and 744 testing pairs.

1) Implementation Details: The proposed model is built on the PyTorch deep learning framework with a hardware configuration of a single NVIDIA RTX A6000 GPU. During training, the Adam optimizer is adopted, with an initial learning rate of 1e-4, a weight decay coefficient of 1e-5, a batch size of 64, and a total of 100 training epochs.

2) Evaluation Metrics: For the text generation task, mainstream metrics such as BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, and CIDEr are employed to measure the semantic consistency and fluency between the generated text and the reference text. In addition, we adopt the SM metric to evaluate the comprehensive performance of the model, which is the average value of BLEU-4, METEOR, ROUGE-L, and CIDEr.

B. Comparison With Existing Methods

The experimental results on the LEVIR-CC and WHU-CDC datasets demonstrate that the proposed model consistently outperforms representative methods. As illustrated in Table I, on LEVIR-CC, which encompasses diverse remote sensing change scenarios, the model exhibits notable advantages in n-gram matching metrics. Specifically, it surpasses SAGE-CC by 0.72% in BLEU-1 and 0.63% in BLEU-2. This improvement stems primarily from the multipath state space network-guided spatial representation module, which efficiently captures long-range change correlations with linear complexity. By circumventing the quadratic computational overhead of conventional Transformers, this module enables precise semantic alignment between change features and descriptive text, thereby enhancing local word order fidelity and phrase-level matching with reference captions. The model also achieves leading performance on comprehensive semantic evaluation metrics. Compared with CTMTNet, it improves METEOR and ROUGE-L by 0.81% and 1.14%, respectively. This gain is largely attributable to the salient change-aware adaptive feature refinement module, which dynamically suppresses background clutter such as nonchanging vegetation and bare land while amplifying salient change signals. As a result, the model focuses more effectively on meaningful change semantics during caption generation, improving both synonym flexibility and long-sequence coherence. Notably, against SEN, which emphasizes change-type semantics, the proposed model achieves a CIDEr score of 138.34, surpassing SEN by 13.32%. This substantial margin arises from the integration of CNN-derived spatial position cues into CLIP semantic features via the bitemporal spatial enhancement guidance unit. This design endows the model with reliable spatial awareness, addressing SEN's limitation in localizing changes and enabling more accurate descriptions of spatial relationships between change regions and surrounding ground objects.

As illustrated in Table II, on the WHU-CDC dataset, which focuses on complex urban changes, the model maintains significant advantages. Its BLEU-4 score exceeds that of SAGE-CC by 2.88%, and its METEOR score surpasses CTMTNet by 3.52%. Moreover, the model achieves a CIDEr score of 160.09, 3.88% higher than SAGE-CC, confirming that its captions are more semantically aligned with the actual change content. The SM score further attests to this superiority, 2.72% above SAGE-CC. Consistent performance gains

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TABLE I

COMPARISONS WITH STATE-OF-THE-ART RSICC METHODS ON THE LEVIR-CC DATASET. BOLD VALUES INDICATE THE BEST PERFORMANCE, WHILE UNDERLINED VALUES REPRESENT THE SECOND-BEST RESULTS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SM
Capt-Diff [38]	72.90	61.98	53.62	47.41	34.47	65.64	110.57	64.52
Capt-Rep [38]	77.64	67.40	59.24	53.15	36.58	69.73	121.22	70.17
Capt-Att-Dual-Att [38]	79.51	67.23	57.46	53.56	37.16	69.19	124.42	71.08
DUDA [38]	81.44	72.22	64.67	57.79	37.15	71.04	124.32	72.58
MCCFormer-S [25]	79.90	70.26	62.68	56.36	39.60	69.46	120.39	71.45
MCCFormer-D [25]	80.42	70.87	62.86	56.38	39.91	70.44	124.44	72.79
RSICCFormer-C [17]	83.09	74.32	66.66	62.41	38.70	73.60	132.62	76.83
PSNet [39]	83.86	75.13	67.89	62.11	38.80	73.60	132.62	76.78
Chg2Cap [19]	84.43	76.35	69.12	62.98	39.42	74.34	136.25	78.25
SEN [28]	85.10	77.05	70.01	64.09	39.59	71.50	125.02	75.05
Diffusion-RSCC [40]	-	-	-	60.90	37.80	71.50	125.60	73.95
CTMTNet [18]	85.95	77.99	70.74	64.69	39.49	74.54	134.94	78.42
PromptCC [34]	83.66	75.73	69.10	63.54	38.82	73.72	136.44	78.13
SAGE-CC [22]	85.69	77.85	71.03	65.50	39.92	74.77	137.50	79.42
Ours	**86.41**	**78.48**	**71.36**	**65.57**	**40.30**	**75.68**	**138.34**	**79.97**

TABLE II

COMPARISONS WITH STATE-OF-THE-ART RSICC METHODS ON THE WHU-CDC DATASET

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SM
DUDA [38]	79.04	69.53	61.57	55.64	34.29	68.98	121.85	70.19
MCCFormer-S [25]	82.14	76.29	71.08	66.51	43.50	79.76	148.88	84.66
MCCFormer-D [25]	73.29	67.88	64.03	60.96	39.69	73.67	134.92	77.31
RSICCFormer-C [17]	78.25	72.82	68.57	65.14	44.35	76.50	143.44	82.36
MaskApproxNet [41]	81.34	75.68	71.16	67.73	43.89	75.41	135.31	80.59
CTMTNet [18]	83.56	77.66	72.76	69.00	45.99	79.23	149.40	85.91
CTM [42]	85.36	79.49	75.36	72.36	46.38	80.27	153.29	88.08
SAGE-CC [22]	86.04	81.16	77.28	74.42	47.89	80.90	156.21	89.86
Ours	**88.69**	**84.06**	**80.20**	**77.30**	**49.51**	**83.43**	**160.09**	**92.58**

TABLE III

COMPUTATIONAL COMPLEXITY FLOPS COMPARISON WITH REPRESENTATIVE RSICC METHODS

Method	**MCC-S**	**MCC-D**	**RSICC**	**PSNet**	**PromptCC**	**Ours**
FLOPs (G)	25.09	25.09	27.10	**13.78**	19.88	18.29

across two datasets validate the robustness and generalization capability of the proposed framework.

As shown in Table III, our SACNet achieves competitive inference floating-point operations (FLOPs), significantly lower than representative Transformer methods such as MCCFormer-S/D and RSICCFormer, highlighting its superior computational efficiency. Compared with PSNet, SACNet only incurs a 32.8% increase in FLOPs while achieving a notable BLEU-4 improvement, thus realizing a better performance-efficiency tradeoff. It also outperforms PromptCC comprehensively with lower FLOPs, further validating the effectiveness of the architecture for RSICC tasks.

C. Ablation Study

Ablation studies on LEVIR-CC and WHU-CDC (see Tables IV and V) yield three key insights.

1) Effect of SGSR: The SGSR unit is essential for achieving precise semantic-spatial alignment in RSICC. The variant without SGSR feeding CLIP extracted features directly into CAFCM and paired with MSRM exhibits a 0.84% lower BLEU-1 and 1.16% lower METEOR on LEVIR-CC compared to the full model. On WHU-CDC, its CIDEr score is 1.42% lower, confirming CLIP-derived semantic features combined with CAFCM background suppression can partially mitigate spatial ambiguity but rely on SGSR to achieve accurate alignment between descriptive semantics and change regions.

2) Effect of CAFCM: The change-driven adaptive feature calibration module plays a critical role in enhancing the fidelity of change-related descriptions by suppressing redundant background clutter and strengthening discriminative change signals. The variant without CAFCM SGSR paired with MSRM shows a 0.60% lower BLEU-4 and 3.65% lower CIDEr on LEVIR-CC relative to the full model. On WHU-CDC,

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TABLE IV
ABLATION EXPERIMENT RESULTS ON LEVIR-CC DATASET

Method			Metrics							
SGSR	CAFCM	MSRM	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr	SM
×	×	×	84.99	76.55	69.07	62.91	39.35	73.68	133.18	77.10
×	✓	✓	85.57	78.22	**71.46**	**65.82**	39.14	74.34	133.90	78.30
✓	✓	×	85.96	77.80	70.57	64.47	39.87	74.92	136.75	79.00
✓	×	✓	85.85	77.94	70.90	64.97	39.50	74.45	134.69	78.40
✓	✓	✓	**86.41**	**78.48**	71.36	65.57	**40.30**	**75.68**	**138.34**	**79.97**

TABLE V
ABLATION EXPERIMENT RESULTS ON WHU-CDC DATASET

Method			Metrics							
SGSR	CAFCM	MSRM	BLEU-1	BLEU-2	BLEU-3	BLEU-4	Meteor	ROUGE-L	CIDEr	SM
×	×	×	86.09	80.83	76.30	72.92	47.60	80.92	153.85	86.93
×	✓	✓	88.68	83.95	79.80	76.70	48.83	82.81	158.67	91.75
✓	✓	×	87.83	83.08	78.76	75.62	48.29	82.06	155.85	90.46
✓	×	✓	88.47	83.83	79.73	76.74	49.26	83.25	159.81	92.27
✓	✓	✓	**88.69**	**84.06**	**80.20**	**77.30**	**49.51**	**83.43**	**160.09**	**92.58**

TABLE VI
ABLATION EXPERIMENT RESULTS ON LEVIR-CC DATASET. PATH DEFINITIONS: PATH1 = HORIZONTAL SCANNING, PATH2 = VERTICAL SCANNING, PATH3 = REVERSE HORIZONTAL SCANNING, PATH4 = DIAGONAL SCANNING

MSRM Spatial Scanning Paths				Captioning Metrics (%)							
Path1	Path2	Path3	Path4	BLEU-1	BLEU-2	BLEU-3	BLEU-4	Meteor	ROUGE-L	CIDEr	SM
×	✓	✓	✓	86.58	78.41	71.17	65.19	**40.35**	75.37	137.05	79.49
✓	×	✓	✓	84.95	77.29	70.38	64.87	39.01	73.40	132.09	77.34
✓	✓	×	✓	**86.60**	**78.94**	**71.89**	66.03	40.22	**75.71**	136.63	79.65
✓	✓	✓	×	85.37	78.31	71.81	**66.16**	39.24	74.72	133.06	78.30
✓	✓	✓	✓	86.41	78.48	71.36	65.57	40.30	75.68	**138.34**	**79.97**

its METEOR decreases by 0.25% and SM by 0.31%, demonstrating CAFCM effectively suppresses redundant background clutter and strengthens discriminative change signals, essential for improving the semantic accuracy of generated captions.

3) Effect of MSRM: The multipath SSM-guided spatial representation module enables robust modeling of diverse spatial change topologies with linear computational complexity. The variant without MSRM SGSR paired with CAFCM achieves a 0.97% lower SM on LEVIR-CC, while on WHU-CDC, its BLEU-4 and CIDEr are 1.68% and 4.24% lower than the full model, respectively. Notably, on WHU-CDC complex urban scenes, the variant with SGSR and MSRM but without CAFCM outperforms the variant with CAFCM and MSRM, validating MSRM's unique capability to model long-range spatial structures such as roads and building clusters with linear computational complexity, a strength lacking in conventional sequential modeling methods. Notably, the full model outperforms any individual component across most metrics on both datasets, which proves that the integration of SGSR, CAFCM, and MSRM enables synergistic collaboration to generate high-quality remote sensing image change description texts.

Table VI presents pathwise ablation results of the multipath SSM-guided module on the LEVIR-CC dataset. Removing any single scanning path leads to performance degradation in key metrics. Omitting Path2 results in the lowest BLEU-1 and CIDEr. Excluding Path4 reduces ROUGE-L to 74.72%. Although individual path removals yield marginal gains in specific metrics, the full model with all four paths outperforms all ablated variants. It achieves the best CIDEr and SM scores. This confirms that integrating diverse spatial traversal patterns is critical for capturing complex change topologies and ensuring comprehensive long-range dependence modeling. Meanwhile, we supply the hyperparameter analysis experiments of the model regarding the number of layers n in the MSRM and beam size in the Supplemental Material.

D. Visual Results

The visualization in Fig. 3 illustrates the region change captioning performance of our model on the LEVIR-CC and

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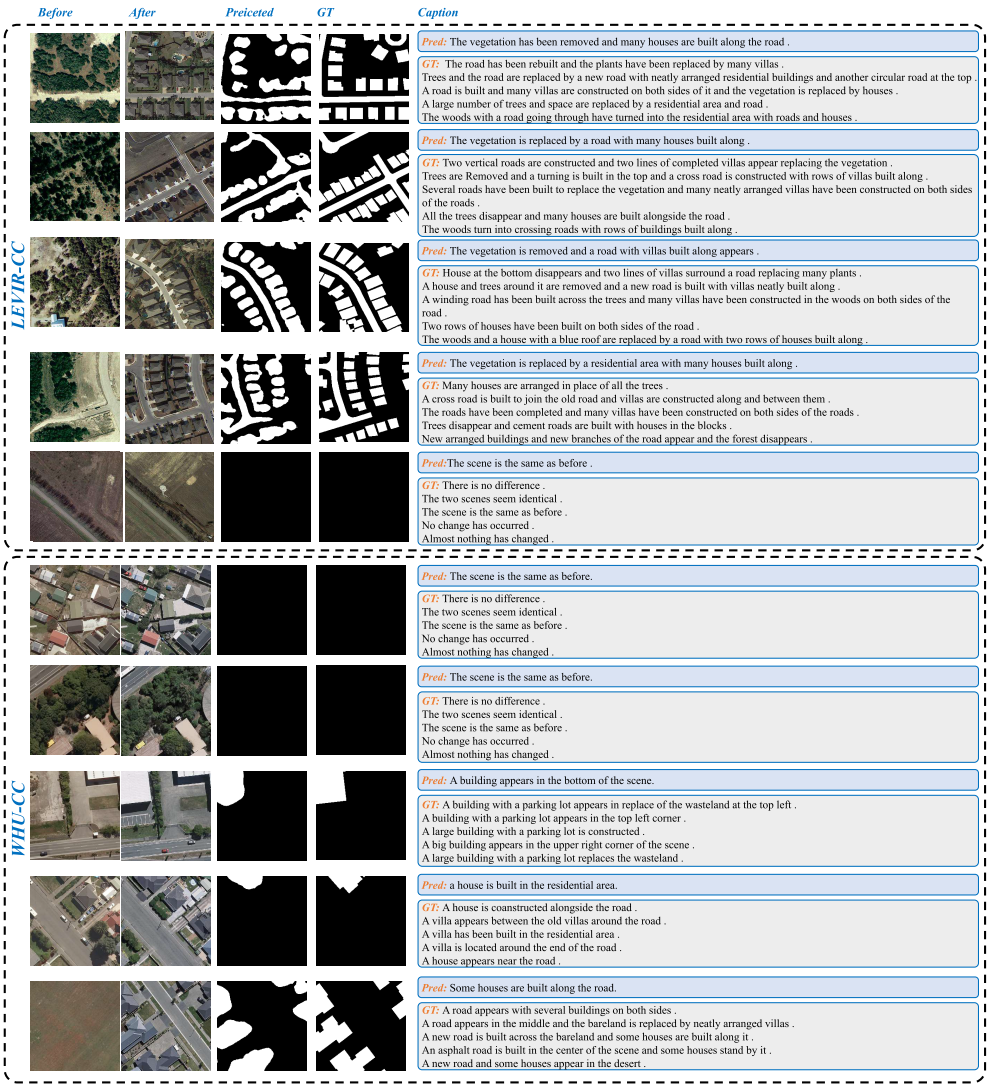


Fig. 3. Region change captioning results on LEVIR-CC and WHU-CDC datasets: each group pairs satellite views with prediction and GT annotation maps, alongside captions describing the regional transformations.

WHU-CDC datasets. As shown in the upper panels corresponding to LEVIR-CC, the dataset primarily focuses on large-scale transition scenarios where vegetation is converted to built-up areas. In the first row of LEVIR-CC, the model's predictions accurately capture the core change event: vegetation removal and subsequent house construction along roads. These predictions align closely with the key transformation described in the GT. This consistency is further demonstrated in the third row of LEVIR-CC: the predictions reflect the replacement of vegetation with residential areas and road extensions, perfectly matching the GT's emphasis on these critical scene modifications. The lower panels correspond to the WHU-CDC dataset, which covers unchanged scenes and local small-scale modifications. Predictions for unchanged scenarios align fully with GT. For local small-scale changes, the model accurately identifies core events, including new building construction with parking lots, villa addition in residential areas, and new road construction with adjacent house erection. These identifications match key transformation details in GT, such as new buildings replacing wasteland, villas appearing among existing residential clusters, and roads extending across bareland with accompanying structures. These results confirm that the model effectively captures both large-scale regional evolution and small-scale local changes. Predictions maintain strong alignment with GT regarding core change semantics.

V. Conclusion

In this article, we alleviate the challenging task of remote sensing image change captioning, focusing on the critical issues of background interference and the effective modeling of diverse spatial change patterns. We propose a novel framework, SACNet, which systematically tackles these challenges through three innovative components. The semantic spatial collaborative encoding module effectively fuses local spatial details from a CNN with global semantics from CLIP, which achieves precise semantic-spatial alignment. The change-driven adaptive feature calibration module dynamically enhances discriminative change features while suppressing irrelevant background information. Notably, the multipath SSM-guided spatial representation module pioneers the use

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of a multipath SSM architecture in RSICC, enabling efficient and powerful modeling of long-range dependencies across various spatial configurations with only linear computational complexity. Comprehensive experiments on the LEVIR-CC and WHU-CDC benchmarks validate the superiority of our approach. SACNet consistently outperforms previous state-of-the-art methods across all standard metrics, demonstrating its capability to generate accurate and semantically rich change descriptions. In the future, we will explore extending this framework to handle temporal sequences with more than two images for more complex change analysis.

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